

DBMS & SQL Mastery Study Guide

Gate Smashers Notes: Commands, Operations, Constraints & Subqueries

1. SQL Command Categorization (DDL vs DML)

Core Taxonomy

DDL (Data Definition Language)

Structure

Defines, alters, and manages database schema structures.

- **CREATE:** Builds a new table or schema.
- **ALTER:** Modifies existing table structure.
- **DROP:** Permanently removes table & structure.
- **TRUNCATE:** Removes all rows; retains structure.

DML (Data Manipulation Language)

Data Rows

Manages and manipulates actual data records inside tables.

- **INSERT:** Adds new data rows.
- **UPDATE:** Modifies existing row values.
- **DELETE:** Removes specified rows (supports WHERE).

2. Structural & Data Operations

Syntax & Comparison

CREATE TABLE Syntax

```
CREATE TABLE employee ( id INT, name  
VARCHAR(20), address VARCHAR(30) );
```

ALTER vs UPDATE

```
-- ALTER: Changes Table Schema (DDL)  
ALTER TABLE employee ADD salary INT; --  
UPDATE: Changes Data Rows (DML) UPDATE  
employee SET salary = salary * 2 WHERE id =  
1;
```

The Data Removal Matrix: DELETE vs TRUNCATE vs DROP

Command	Type	WHERE Clause?	Action & Structure Impact
DELETE	DML	Yes	Deletes specified rows. Table structure and remaining data stay intact.
TRUNCATE	DDL	No	Deletes ALL rows instantly. Table structure is preserved.
DROP	DDL	No	Destroys the entire table (data + schema completely removed).

3. SQL Integrity Constraints

Data Validation

Constraint	Core Purpose & Behavior Rule
PRIMARY KEY	Uniquely identifies each record. Automatically enforces UNIQUE + NOT NULL .
FOREIGN KEY	Maintains referential integrity between parent and child tables.
UNIQUE	Ensures all values in a column are distinct (prevents duplicates).
NOT NULL	Ensures that a column cannot store empty or NULL values.
CHECK	Restricts the value range placed in a column according to a logical condition.
DEFAULT	Provides a predefined fallback value when no explicit value is provided during INSERT.

4. Advanced Subquery Patterns

Nested Logic

💡 Subquery Fundamentals

A **subquery** is a query nested inside another query. The *inner query* executes first, and its result is evaluated by the *outer query*.

1. Find Maximum Salary

```
SELECT MAX(salary) FROM employee;
```

2. Find Employee with Maximum Salary

```
SELECT ename FROM employee WHERE  
salary = ( SELECT MAX(salary) FROM employee  
);
```

3. Find Second-Highest Salary

```
SELECT MAX(salary) FROM employee  
WHERE salary != ( SELECT MAX(salary) FROM  
employee );
```

4. Employee with Second-Highest Salary

```
SELECT ename FROM employee WHERE  
salary = ( SELECT MAX(salary) FROM employee  
WHERE salary != ( SELECT MAX(salary) FROM  
employee ) );
```

5. Aggregation with GROUP BY Pattern

Data Summarization

The **GROUP BY** clause groups rows that share identical values in specified columns into summary rows. Aggregate functions (COUNT, SUM, AVG, MAX, MIN) then operate on each isolated group.

Query: Count Employees per Department

```
SELECT department, COUNT(*) FROM  
employee GROUP BY department;
```

Execution Visual Example

Department	COUNT(*)
HR	2
Marketing	2
IT	1

Key Idea: Aggregate functions process individual groups formed by **GROUP BY**, rather than evaluating the whole table at once.

6. Exam Revision Summary

Quick Recall

DDL vs DML

DDL = Schema & Structure.
DML = Data Rows & Tuples.

Deletion Rule

DELETE = Selective rows.
TRUNCATE = All rows.
DROP = Table schema.

Subquery Logic

Nth Highest = Filter top (N-1) max values, then find MAX.